Scientific Calculator

A desktop calculator built with Java Swing

About

This project is a desktop scientific calculator written in Java using the Swing GUI toolkit. It has a dark theme with a large display at the top and a grid of buttons below, arranged with **GridBagLayout**. The **C** and **=** buttons are red, the **=** button spans two rows, and the **0** button spans three columns.

Features

- Basic arithmetic: addition, subtraction, multiplication, division and decimal numbers
- Brackets and percentage, with correct operator precedence (for example, 2+3*4 gives 14)
- Powers and roots: x^2, x^3, y^x, 2^x, square root and cube root
- Logarithm (base 10), reciprocal (1/x), factorial (x!) and sign change (+/-)
- Trigonometric functions (sin, cos, tan) and hyperbolic functions (sinh, cosh, tanh)
- Rad/Deg toggle for the angle mode
- Quick buttons for pi and a random number (Rnd)
- Error handling: invalid input such as division by zero shows "Error" instead of crashing

Requirements

- Java Development Kit (JDK) 8 or higher
- No external libraries needed (Swing and AWT are part of the JDK)

How to Run

Save the file as **ScientificCalculator.java**, then open a terminal in that folder and run:

```
javac ScientificCalculator.java
java ScientificCalculator
```

How to Use

- Click the number and operator buttons to build an expression on the display, then press **=** to get the result.
- Single-input buttons (sin, sqrt, x^2, 1/x, x! and so on) work on the current display value straight away.
- For powers, enter the base, press **y^x**, enter the exponent, then press **=**.
- Press **Rad** to switch to degrees (the button changes to **Deg**). Press it again to go back to radians.
- Press **C** to clear the display.

Button Reference

Button	What it does
1/x	Reciprocal of the current value
x^2, x^3	Square and cube
y^x	Raises a number to a power (use between two numbers)
2^x	2 raised to the current value
sqrt, 2-root	Square root and cube root
log	Logarithm, base 10
x!	Factorial (whole numbers from 0 to 170)
sin, cos, tan	Trigonometric functions (radians or degrees)
sinh, cosh, tanh	Hyperbolic functions
%	Divides the number before it by 100
+/-	Changes the sign of the current value
pi, Rnd	Inserts pi or a random number between 0 and 1

How It Works

Button presses build an expression string that is shown on the display. When **=** is pressed, a small recursive-descent parser evaluates it. The parser handles precedence in this order: addition and subtraction, then multiplication and division, then powers, then brackets and numbers. Results that are whole numbers are shown without a trailing ".0".

Project Structure

```
ScientificCalculator/
|-- ScientificCalculator.java   (all the code)
`-- README.pdf
```

Technologies Used

- Java
- Swing (JFrame, JPanel, JButton, JLabel)
- AWT (layouts, colors, events)

Known Limitations

- Expressions are built as text, so there is no cursor or backspace button.
- Results are shown as double values, so very long decimals may lose some precision.